

Fig. 2: An actual-size PCB layout of the low-cost 6-bit DAC

mon configuration built around step-down transformer X1 (secondary voltage 18V to 20V with current 1A or above), bridge rectifier BR1 and voltage regulator IC 7805 (IC1). The mains power is applied on connector CON1. 5V is available at connector CON5.

The selection of mains transformer X1, bridge rectifier and heat sinks for IC1 and IC2 depends on the required maximum output current from the DAC. IC1 and IC2 can be mounted on a common heat-sink after proper mounting is done.

The load is connected to connector CON3.

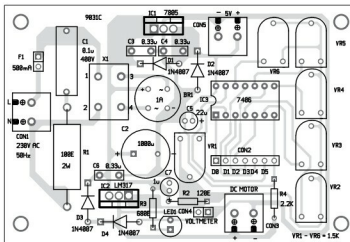


Fig. 3: Component layout of the PCB

through D5 is buffered with 7406 or similar (IC3). With an open collector, the IC works as a translator/buffer between standard TTL levels to higher voltages needed for LM317.

If the requirement of current is more than 1A, then select adjustable regulator IC2 from series LM317T (1.5A), LM350 (3A) or any compatible adjustable-linear regulator.

This makes the DAC adaptable to a lot of applications. In many cases, there is no need to start the output voltage from 0V. This makes the solution even simpler.

Input digital code D0 through D5 is buffered with IC3, which should be obligatory with open connector. The preferred device is 7406 or better, with outputs that can work with up to 30V.

Power requirements are from a com-

A DC voltmeter with 50V range is connected at CON4 for measuring the output voltage. The DAC can be tested with 12V/5W/0.4A light bulb, 12V/0.3A fan, heating element for thermostat with nominal current up to 0.3A and maximum current below 1A and similar loads.

Construction and testing

An actual-size, single-side PCB layout for the low-cost 6-bit DAC is shown in Fig. 2 and its component layout in Fig. 3. •

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